FlowLens / Process intelligence

V.S.B. Engineering College | AI & Data Science
Institutional Process Bottleneck Mining & Delay Analytics

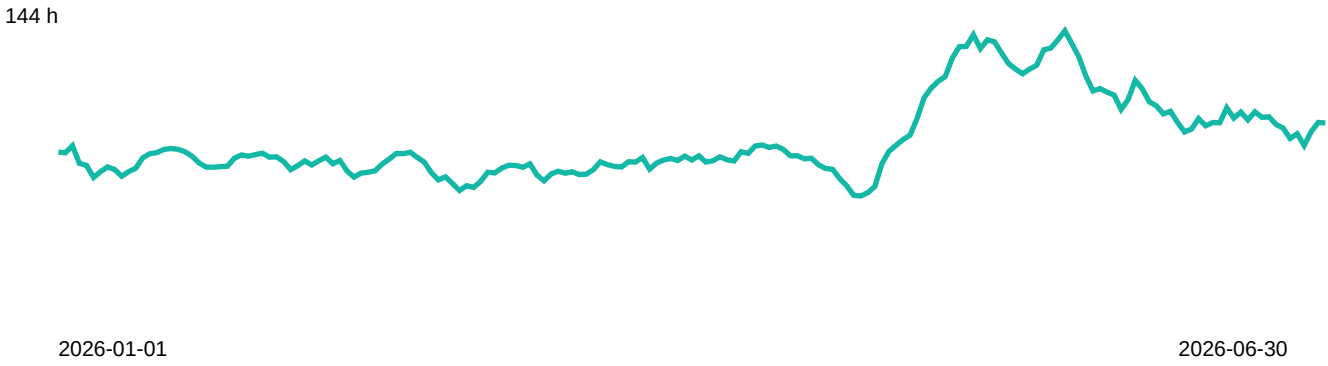
Campus operations · Jan–Jun 2026

Arrival window: 2026-01-01 to 2026-06-30 | Queue clock: elapsed hours

Cases	Mean cycle	SLA breach	Open cases
3,200	87.7 h	15.4%	110

Where requests wait

Arrival-cohort cycle trend (hours)



Prioritised action plan

01. Add a second signing window

Introduce a Monday signing window alongside the weekly batch.

90.0h mean queue · 156.3h p90 · 1,257 cases · 48h target

50% queue reduction scenario: 56,069 total hours saved; 235 SLA breaches avoided. Effort: Medium.

02. Delegate routine HOD approvals

Assign a deputy during exam weeks and send a reminder after 24 hours.

43.6h mean queue · 88.5h p90 · 2,527 cases · 24h target

50% queue reduction scenario: 54,034 total hours saved; 276 SLA breaches avoided. Effort: Medium.

03. Shorten Issued queues

Introduce queue reminders and review capacity at the busiest hours.

4.6h mean queue · 9.8h p90 · 2,970 cases · 24h target

50% queue reduction scenario: 6,820 total hours saved; 14 SLA breaches avoided. Effort: Low.

04. Shorten Staff Verification queues

Introduce queue reminders and review capacity at the busiest hours.

4.6h mean queue · 9.7h p90 · 2,542 cases · 24h target

50% queue reduction scenario: 5,745 total hours saved; 26 SLA breaches avoided. Effort: Low.

05. Balance the office queue

Route new requests to the least-loaded clerk and standardise the document checklist.

4.6h mean queue · 9.5h p90 · 1,892 cases · 12h target

50% queue reduction scenario: 4,544 total hours saved; 6 SLA breaches avoided. Effort: Low.

Method & limitations

Queue time is the gap between one activity ending and the next starting. Service time is measured separately. Open cases are excluded from SLA rates. Scenario estimates hold other waits fixed and do not model capacity feedback. Associations are not proof of causation. Sample datasets are synthetic.